

When Does Driver Agency Matter? Comparing Data Replay and Agent-Based Optimisation for Paratransit Electrification in Cape Town, South Africa

J. T. Sello^{a,*}, M. J. Booysen^{a,b,c}

^a*Electrical and Electronic Engineering, Stellenbosch University, South Africa*

^b*Industrial Engineering, Stellenbosch University, South Africa*

^c*Department of Computing and Technology, Uganda Christian University, Uganda*

Abstract

Complete abstract.

Keywords: paratransit, minibus taxi, electrification, agent-based modelling, data replay, charging infrastructure, simulation-based optimisation, Cape Town

1. Introduction

TODO: Complete introduction.

1.1. Paratransit electrification as a coupled planning problem

1.2. Modelling paratransit: the tale of two paradigms

1.3. Research gap

Existing work can evaluate whether observed paratransit mobility is compatible with electrification [], while behaviourally explicit models can represent adaptive responses to energy and charging constraints []. What remains unclear is whether these different representations lead to materially different infrastructure planning decisions when applied to the same empirically observed transport subsystem, and whether infrastructure selected under fixed-mobility assumptions remains effective once driver and passenger interactions are allowed to respond endogenously.

1.4. Research questions

This study addresses the following research questions:

1. How do predicted electrification outcomes differ when the mobility of the same sampled paratransit fleet is represented as fixed empirical activity versus endogenous driver behaviour?
2. How do the electrification infrastructure strategies selected by data replay and agent-based optimisation differ when each model is evaluated using the outcomes it can validly represent?
3. What additional operational and passenger-service consequences become visible when a replay-derived infrastructure strategy is evaluated in the agent-based model?

1.5. Contributions

This study examines the implications of explicit behavioural representations of driver agency for paratransit electrification planning. It contributes to the literature by performing a controlled comparison of fixed-mobility data-replay and decision-adaptive agent-based modelling approaches applied to the same empirically observed paratransit subsystem. The model-comparison framework adheres to model-native objective spaces, rather than forcing a single objective space reduction across models with different behavioural assumptions.

*Corresponding author

Email addresses: jtsello@sun.ac.za (J. T. Sello), mjbooyesen@sun.ac.za (M. J. Booysen)

Additionally, the study performs independent simulation-based optimisation of charging infrastructure under each model, and evaluates the consequences of applying a data-replay-derived infrastructure strategy in the agent-based model. The findings provide insights into how progressively richer representations of driver agency can influence electrification outcomes and infrastructure planning decisions, guiding future research and policy development in the context of paratransit electrification.

2. Related work

TODO: Complete literature review.

3. Methodology

TODO: Complete methodology.

3.1. Study design

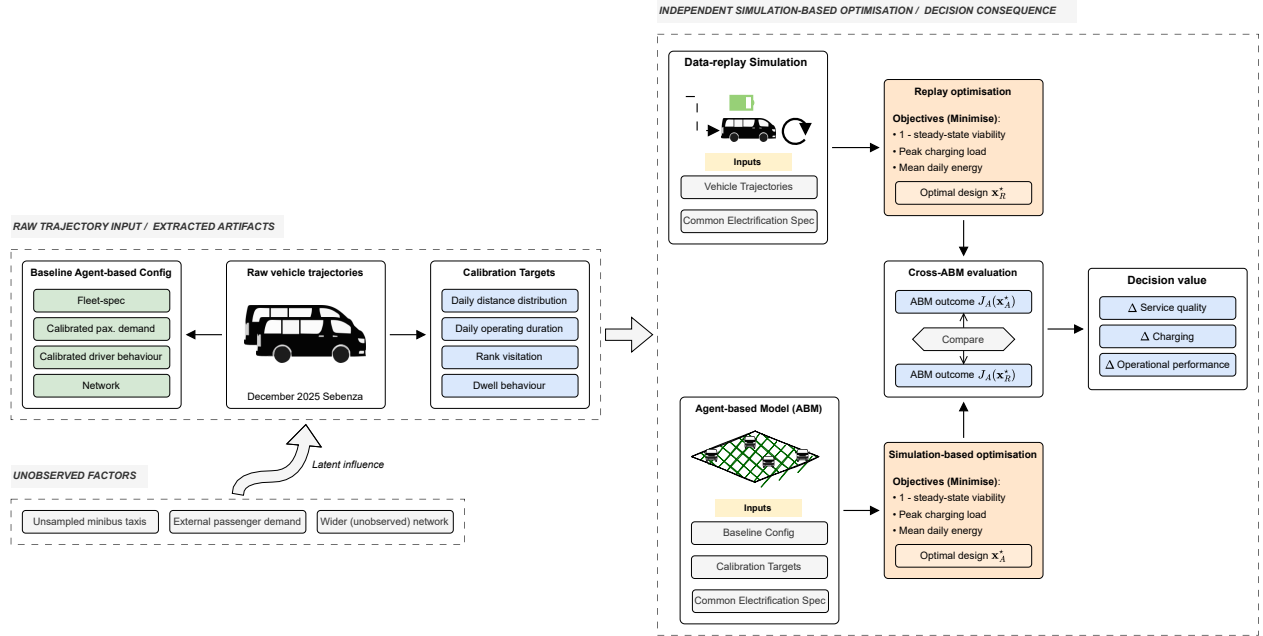


Figure 1: Overview of the study design. The same empirical paratransit mobility data is used to parameterise both a data-replay model and an agent-based model. Each model is then used to independently optimise charging infrastructure, and the resulting infrastructure strategies are evaluated under each model’s native assumptions. The agent-based model is also used to evaluate the consequences of applying a data-replay-derived infrastructure strategy in a decision-adaptive context. The result is then compared to the agent-based model’s native infrastructure strategy to assess the implications of ignoring driver agency in electrification planning.

4. Results

TODO: Complete results.

4.1. Empirical model validation

4.2. Baseline electrification under data replay

4.3.

5. Discussion

TODO: Complete discussion.

Table 1: Cumulative recovery of independent City of Johannesburg taxi-rank references through the reconstruction stages.

Metric	M_0	M_1	M_2
Daily vehicle distance	-		
Operating duration	-		
Rank arrival rate	-		
Rank departure rate	-		
Dwell duration	-		
Move start time	-		
Move end time	-		

Note: Model M_0 represents the mobility observed in the empirical data and is therefore used as the ICE-era reference.

6. Conclusion

TODO: Complete conclusion.

References